

# Sampling distribution



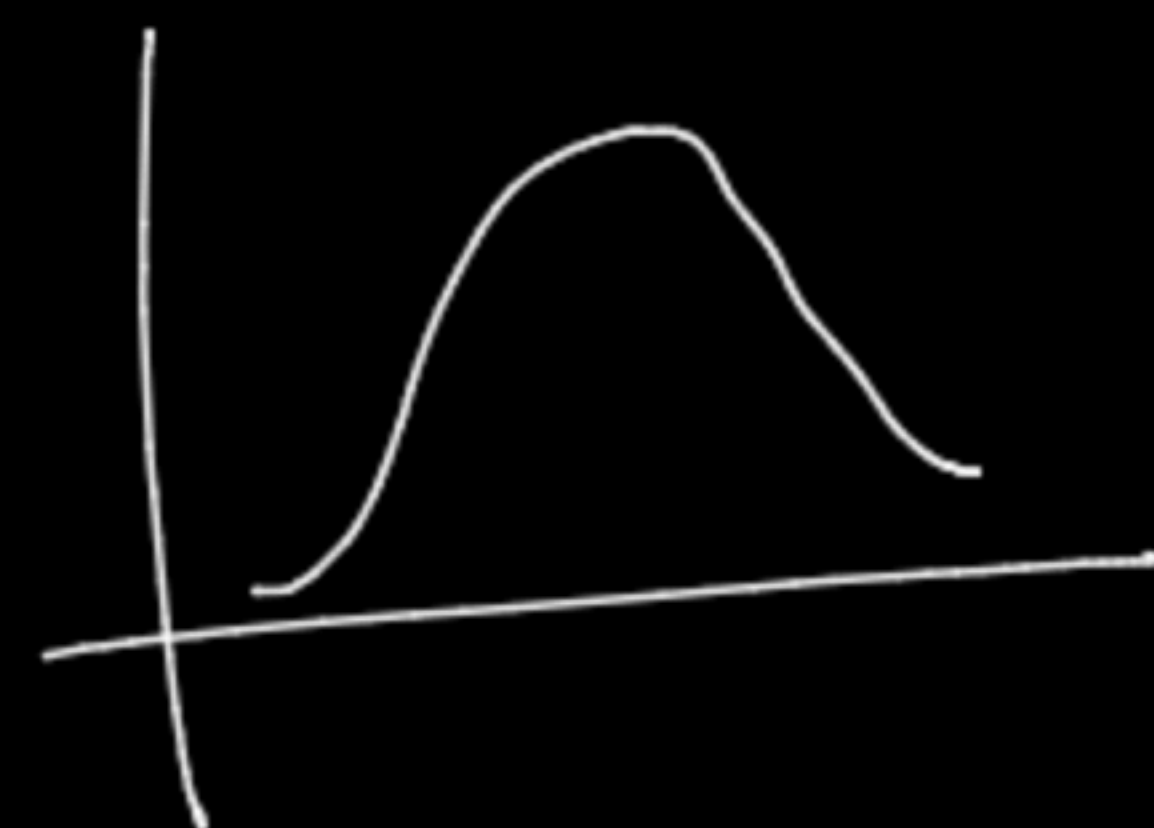
'CLT'  $\rightarrow$   $X \sim \text{Random} \rightarrow \text{samples}$



with  $n=30$   $\rightarrow$   $S_1 \rightarrow \bar{X}_1$   
 $S_2 \rightarrow \bar{X}_2$

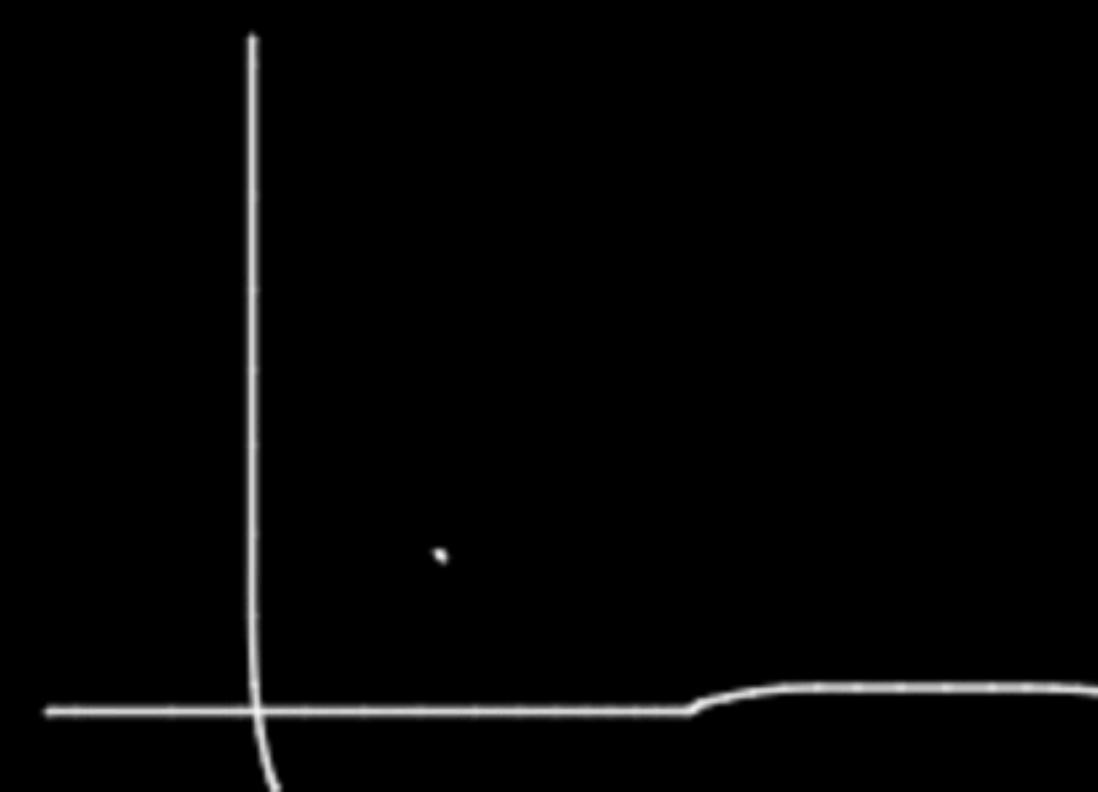
$\vdots$   
 $S_K \quad \bar{X}_K$

$\rightarrow$  plot mean



$$N\left(\mu, \sqrt{\frac{\sigma^2}{n}}\right) \text{ as } \underline{\underline{n \rightarrow \infty}}$$

$\rightarrow$  Gaussian



$\mu$   
 $\uparrow$   
mean  
of your  
population

implement  
 $\uparrow$



Q-Q plot

↑  
Quantile-Quantile  
↑

↓  
 $X \rightarrow x_1 \ x_2 \ x_3 \ \dots \ x_{500}$   
↑

$x \sim N(?, ?)$

How to check → ND

↓

How to plot

(step 1) → sort it  $x_i$  and compute percentile

$x_1 \ x_2 \ x_3 \ x_4 \ \dots \ x_{500}$

↑  
ND on plot  
Yes

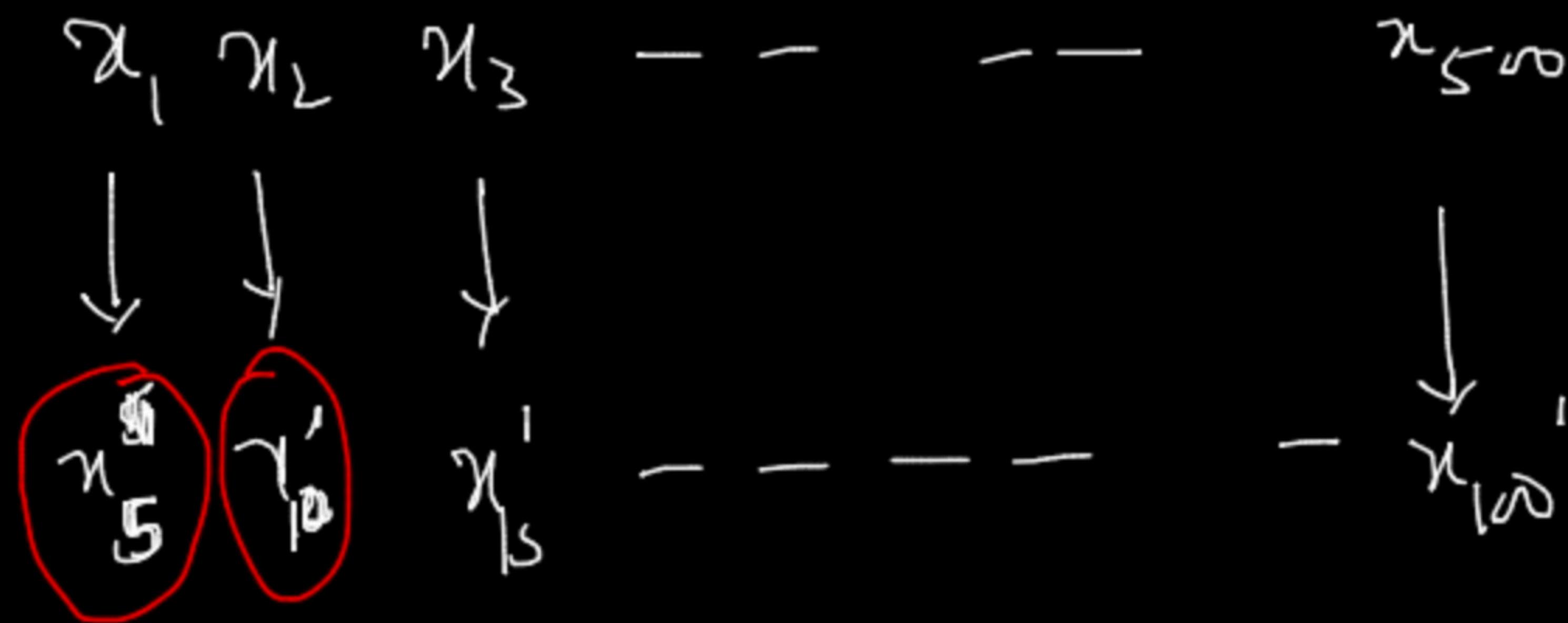
'proof'  
↑

① Q-Q plot

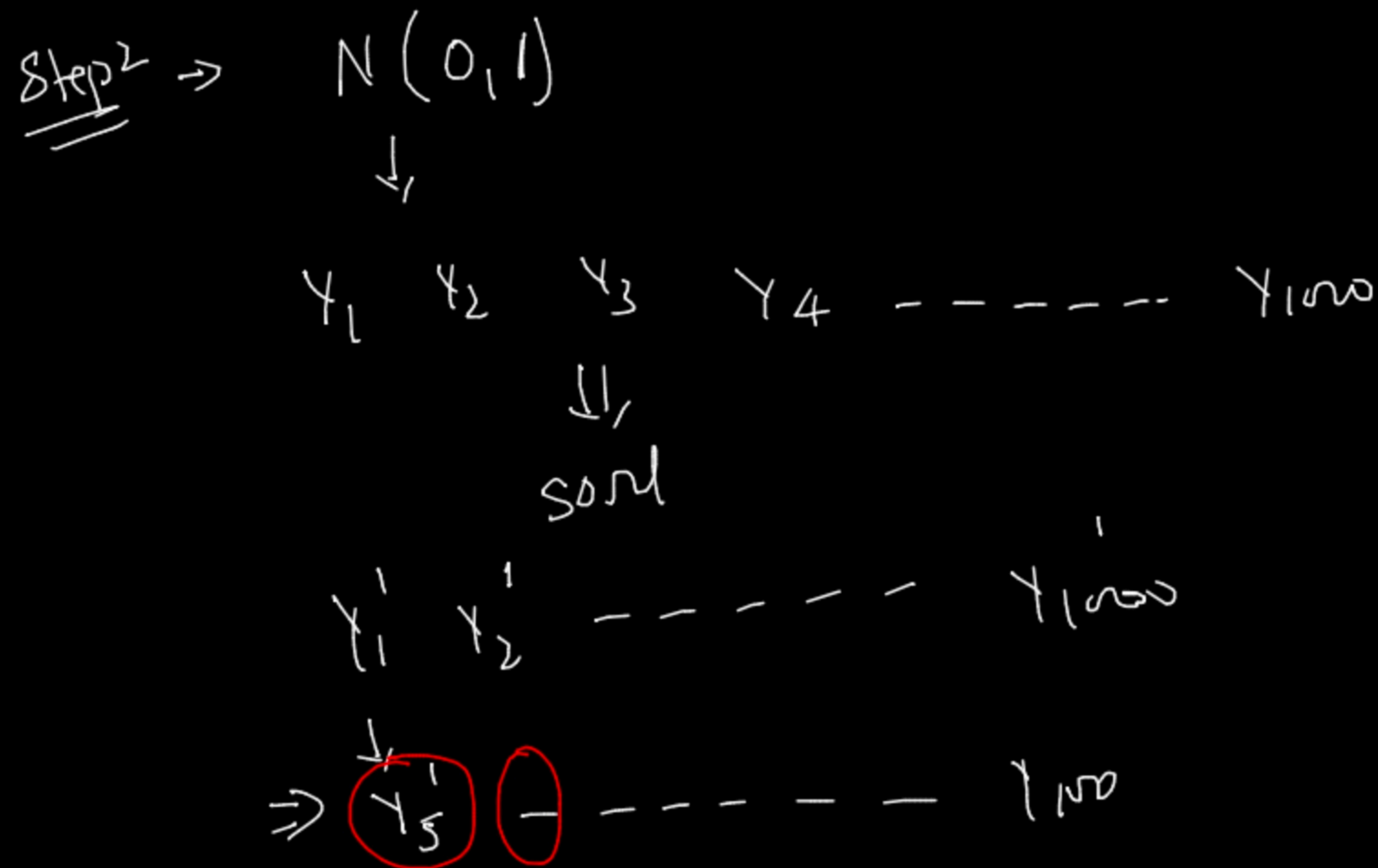
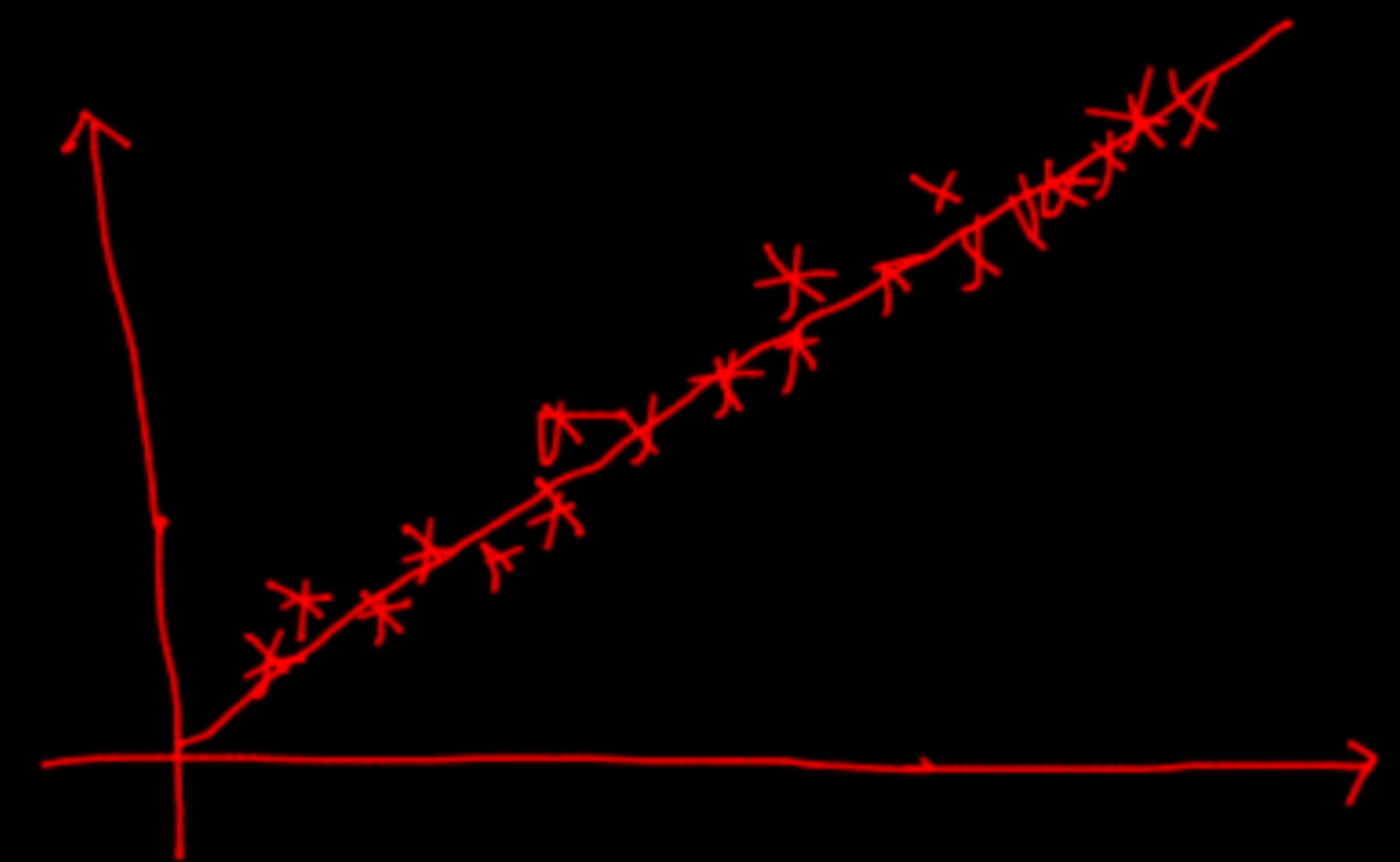
② Statistical testing

{ KS  
AD }





$(x'_5, y'_5)$   
 $(x'_{10}, y'_{10})$





# How/where to use distribution?

XYZ

Task  $\rightarrow$  t-shirts

$\rightarrow$  XL

Q How many XL t-shirts s — on —

Sol<sup>n</sup>

collect data for all emp<sup>s</sup> 100

OR

domely  
K<sub>s</sub> { height  $> 180$ cm  $\rightarrow$  XL  $\rightarrow$  t-shirt  
160cm  $< h < 180 \rightarrow$  L

$$\mu = 150\text{cm}$$

$$\sigma = 10\text{cm}$$

$$\rightarrow -\sigma \rightarrow \underline{\underline{68\%}}$$

$$\begin{aligned} & (\mu - \sigma \leq x \leq \mu + \sigma) \\ & \rightarrow (140, 160) \rightarrow \underline{\underline{68\%}} \end{aligned}$$

$$(\mu - 2\sigma \leq x \leq \mu + 2\sigma)$$

$$[130, 170] \rightarrow \underline{\underline{95\%}}$$

1

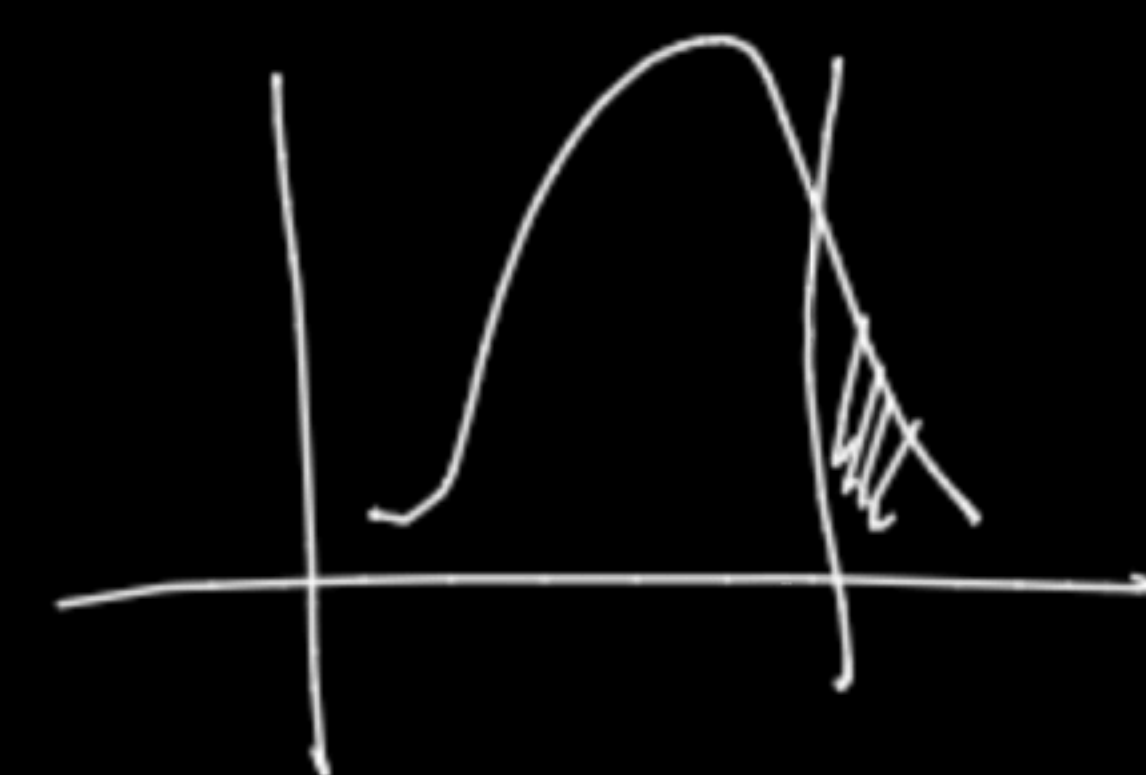
mean  
std

50

mean = ?

std deviation

$$P(X > 180)$$



$$N(\mu, \sigma^2)$$



$\mu, \sigma$

data (x)  $\rightarrow$  any distribution  $\uparrow$   $\boxed{PDX}$

$\frac{x-\mu}{\sigma}$

$\rightarrow \mu - 2\sigma \text{ to } \mu + 2\sigma$

time

$\mu = 150$   
 $\sigma = 10$

$$k\sigma - \mu \leq X \leq \mu + k\sigma$$

$$X - \mu \geq k\sigma$$

$$\boxed{X \geq k\sigma + \mu}$$

OR  $\boxed{X \leq k\sigma - \mu}$

$$[150 - 20, 170]$$

$\downarrow$

$$[130, 170] \rightarrow \frac{x-\mu}{\sigma}$$

(1hr)

$\rightarrow$  'chebyshev's law'

$\downarrow$

$$P(|X - \mu| > k\sigma) \leq \frac{1}{k^2}$$

any numbers  $\uparrow$

$\Rightarrow$

$$P(\mu - k\sigma < X < \mu + k\sigma) > 1 - \frac{1}{k^2}$$



→  $[20K, 60K] \rightarrow ?$

$$\begin{cases} \mu = 40K \\ \sigma = 10K \end{cases}$$

$(20K) \rightarrow \mu - 2\sigma$   
 $\mu = 40K$   
 $40 - (2 \times 10) = 20$

$(60K) \rightarrow \mu + 2\sigma$   
 $40 + 2 \times 10 = 60$

$K = 2$

$P(20 < X < 60) \Rightarrow 1 - \frac{1}{2^2}$   
 $\uparrow$

$P(20 < X < 60) \Rightarrow 0.75$   
 $\uparrow$

Blackboard